SIMATS ENGINEERING

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Saveetha Institute of Medical And Technical Sciences

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ITA0329– MOBILE COMPUTING

1G to 5G Evolution Interactive Learning Platform.

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ABSTRACT

- The 1G to 5G Evolution Interactive Learning Platform is a mobile app that helps students learn how mobile networks changed from 1G to 5G.
- It presents each generation’s data rate, spectrum band, latency, technologies and services through an interactive timeline and charts in a simple, visual way.
- Interactive Timeline & Data Visualisation, and Gamified Quiz & Progress Tracking, which track quiz score, engagement time and comparison accuracy.
- Existing study methods (textbooks, notes, slides) are static with no interaction or feedback, whereas the proposed app is interactive, works offline and tracks learner progress.

INTRODUCTION

- Mobile networks have grown from 1G (voice only) to 5G (very high speed, low delay and smart devices).
- Textbooks explain this evolution in long text and tables, which students find hard to remember.
- This app turns the same content into an interactive timeline, charts and quizzes.
- It makes learning simple and interesting, helps compare generations easily and tests understanding through quizzes.
- It used by students and teachers for telecom and mobile computing courses, self-study and exam revision on a phone, even offline.

PROBLEM STATEMENT

- Students find it hard to remember and compare 1G to 5G because textbooks give no interactive content and no quizzes, instant feedback or progress tracking .

OBJECTIVES

- **Content and Visualization:** To display 1G to 5G information using an interactive timeline and charts, and to make comparing network generations easy.
- **Quiz and Assessment:** To conduct interactive quizzes with instant feedback, and to track quiz scores, learning time and comparison accuracy.

EXISTING SYSTEM

- Students learn from textbooks, notes and slides.
- No interaction or practice for the learner.
- Comparing generations takes a lot of effort.
- No tracking of learner progress.

PROPOSED SYSTEM

- **Interactive Timeline & Data Visualisation:** shows the evolution from 1G to 5G step by step; tapping a generation displays its data rate, spectrum band, latency, technologies and services, and charts compare generations side by side.
- **Gamified Quiz & Progress Tracking:** quizzes with points, levels, badges and instant feedback, with quiz score, engagement time and comparison accuracy saved in a progress dashboard.

MODULES

- **Module 1 – Interactive Timeline & Data Visualisation:** a scrollable 1G to 5G timeline with generation details and comparison charts (MPAndroidChart), using the SQLite content store so it works offline.
- **Module 2 – Gamified Quiz & Progress Tracking:** questions picked by generation or topic, score calculation with instant feedback, and a progress dashboard for each learner.
- **Outcome:** students understand and compare mobile generations easily, practise through quizzes and can see their progress, anytime on a phone.